

MASTER PORTFOLIO

GENOMIC PRIVACY, SECURITY, ACCESSIBILITY, AND PUBLIC POLICY

Production Edition – Five-Volume Architecture

MASTER INDEX

Volume	Title	Primary Audience
I	Constitutional Genomic Privacy	Appellate counsel, jurists, constitutional scholars
II	Genomic Security & Governance	Cybersecurity auditors, systems architects, technical reviewers
III	Disability, Accessibility & Administrative Systems	SSA policy analysts, administrative law practitioners, disability advocates
IV	Scientific Record & Validation	Bioinformaticians, peer reviewers, clinical researchers
V	Legislative and Regulatory Frameworks	Congressional staff, state legislators, regulatory counsel

PORTFOLIO PURPOSE

This portfolio examines whether genomic information constitutes a unique category of personal information requiring heightened legal protection due to its immutable, uniquely identifying, predictive, familial, and persistently sensitive characteristics.

The portfolio further evaluates technical, scientific, administrative, and legislative mechanisms for protecting genomic information while supporting responsible research, healthcare innovation, accessibility, and public administration.

CORE THESIS

Genomic information differs fundamentally from ordinary medical records because it:

- **Cannot be changed or revoked.** (Immutability)
- **Uniquely identifies individuals.** (Unique Identifiability)
- **Reveals information about biological relatives.** (Familial Impact)
- **Predicts future health risks and traits.** (Predictive Capacity)
- **Remains sensitive throughout an individual's lifetime.** (Persistent Sensitivity)

These characteristics may justify legal protections beyond those currently provided by conventional privacy frameworks.

CROSS-REFERENCE SYSTEM

Prefix	Domain	Range
CP	Constitutional References	CP-001 through CP-999
SG	Security References	SG-001 through SG-999
DA	Disability References	DA-001 through DA-999
SR	Scientific References	SR-001 through SR-999
LF	Legislative References	LF-001 through LF-999
EX	Exhibits	EX-0001 through EX-9999

FINAL PRODUCTION REQUIREMENTS

1. Every factual assertion linked to an exhibit.
2. Every exhibit assigned a unique identifier.
3. Separate factual findings from policy recommendations.
4. Separate scientific observations from legal arguments.
5. Separate constitutional claims from legislative proposals.
6. Maintain full chain-of-custody documentation for all records.
7. Maintain version-controlled document history.
8. Include executive summaries for each volume.
9. Include glossary, acronym index, bibliography, and master exhibit register.

VOLUME I: CONSTITUTIONAL GENOMIC PRIVACY

Executive Summary

Volume I argues that genomic information occupies a doctrinal category that existing privacy frameworks do not adequately address. It traces the evolution of Fourth and Fourteenth Amendment privacy jurisprudence, identifies a judicially manageable Question Presented, and proposes a tiered constitutional framework that courts can apply to governmental genomic data practices. The volume does not create new rights; it demonstrates that existing doctrine, properly understood, compels heightened protection for genomic information.

CHAPTER 1: Genomic Information as a Distinct Category of Personal Data

1.1 Purpose

This chapter establishes the factual and conceptual predicate for the constitutional analysis that follows. It defines genomic information, distinguishes it from conventional medical records and biometric identifiers, and identifies five characteristics that render it qualitatively different from every category of personal data previously addressed by constitutional doctrine.

1.2 Definition and Scope

For purposes of this portfolio, **genomic information** means the complete or partial deoxyribonucleic acid (DNA) sequence of an individual, including:

- **Whole-genome sequence (WGS):** The approximately 3.2 billion base pairs comprising the individual's complete nuclear genome.
- **Exome sequence:** The protein-coding subset of the genome, approximately 1–2% of the total.
- **SNP array data:** Genotype information at selected single-nucleotide polymorphism sites, typically ranging from hundreds of thousands to several million loci.
- **Mitochondrial DNA (mtDNA):** The maternally inherited mitochondrial genome.
- **Derived inferences:** Computational predictions, risk scores, trait assessments, or familial relationship determinations generated from the sequence data.

This definition excludes somatic mutations that are not heritable (e.g., cancer mutations confined to tumor tissue) unless such data is linked to germline information in a manner that permits identification or inference about the individual's heritable genome.

1.3 The Five Foundational Characteristics

1.3.1 Immutability (CP-001)

An individual's germline genome is fixed at conception and remains unchanged throughout life, with the trivial exception of accumulated somatic mutations that do not affect the heritable sequence. Unlike passwords, financial account numbers, or even Social Security numbers, genomic information cannot be altered, rotated, or revoked if compromised. Once disclosed, the individual has no remedial mechanism to restore privacy.

Doctrinal significance: Immutability eliminates the standard mitigation strategy in data-breach contexts. It also means that any governmental collection creates a permanent record of biological identity that cannot be escaped through changed circumstances.

1.3.2 Unique Identifiability (CP-002)

A whole-genome sequence is, for all practical purposes, unique to the individual. Even partial profiles—such as the 20 CODIS short tandem repeats or genome-wide SNP arrays—provide sufficient statistical discrimination to identify individuals with near-certainty when compared against reference databases. Moreover, re-identification attacks have demonstrated that supposedly "anonymized" genomic datasets can be linked to named individuals using publicly available genealogical databases, demographic metadata, or surname inference algorithms.

Doctrinal significance: Genomic data functions as a persistent biometric identifier that remains valid across contexts and time. Unlike a temporary identifier (e.g., a session cookie), it does not expire.

1.3.3 Predictive Capacity (CP-003)

Genomic information reveals not only present biological states but probabilistic inferences regarding future health risks, hereditary conditions, biological predispositions, and potentially behavioral or cognitive traits. Polygenic risk scores (PRS), machine-learning inference models, and advancing functional genomics continuously expand the scope of predictions derivable from a static genome.

Doctrinal significance: Predictive capacity transforms genomic data from a historical record into a forward-looking profile. Governmental acquisition of such information implicates liberty interests in avoiding preemptive classification or discrimination based on probabilistic biological inferences rather than actual conduct or manifested conditions.

1.3.4 Familial Impact (CP-004)

Genomic information is inherently relational. Shared inheritance patterns mean that disclosure, analysis, or inference-generation from one individual's genome necessarily reveals information about biological relatives—parents, siblings, offspring, and extended family members—who have not consented to the disclosure. Technical segregation of "individual" from "familial" genomic

information is impossible because the signals of shared inheritance are embedded in the sequence itself.

Doctrinal significance: No other category of personal information routinely produces this collateral exposure as an inherent, unavoidable feature of the data type. The familial dimension means that individual consent cannot fully legitimate governmental access.

1.3.5 Persistent Sensitivity (CP-005)

The significance of genomic information is not static. As scientific knowledge advances, a genome that today reveals limited information may tomorrow reveal substantially more. The individual who discloses a genome for a narrow purpose (e.g., ancestry analysis in 2020) cannot predict the range of inferences that will become possible by 2035. The data carries a "latent sensitivity" that increases over time without any new collection or disclosure by the individual.

Doctrinal significance: Traditional privacy analysis assumes that the sensitivity of disclosed information is fixed at the moment of disclosure. Genomic information violates that assumption, undermining the premise that disclosure to a third party for a limited purpose constitutes a blanket surrender of privacy expectations.

1.4 Distinction from Conventional Categories

Category	Why Genomic Data Exceeds It
Medical record	Medical records document history; genomes predict future states and identify relatives.
Biometric identifier	Fingerprints identify; genomes identify, predict, and implicate relatives.
Physical evidence	Blood or tissue is a source; the extracted information is what persists and propagates.
Digital data	Genomic data is digital in storage but biological, immutable, and lifetime-persistent in content.
Voluntarily disclosed	Genomic disclosure is often medically necessary, not freely chosen; the data cannot be revoked.

CHAPTER 2: Historical Privacy Jurisprudence

2.1 Purpose

This chapter establishes the doctrinal foundation upon which the Proposed Constitutional Framework in Chapter 5 rests. It traces the evolution of Fourth and Fourteenth Amendment privacy doctrine from physical trespass through digital surveillance, demonstrating at each stage why genomic information breaks the analytical categories that courts have developed for conventional records, physical evidence, and even advanced digital data.

2.2 Fourth Amendment Foundations: From Trespass to Reasonable Expectation

2.2.1 The Trespass Doctrine

The original Fourth Amendment framework required a physical intrusion upon constitutionally protected areas—persons, houses, papers, or effects. *Boyd v. United States*, 116 U.S. 616 (1886), established that compelled production of private papers could constitute an unreasonable search. The physical-invasion requirement dominated early doctrine: unless the government crossed a property boundary or seized a tangible object, no "search" occurred.

Doctrinal significance for genomic privacy: The trespass framework is insufficient for genomic data because the privacy harm arises not from physical intrusion upon the body (though that may occur at collection) but from the informational content extracted and its subsequent retention, aggregation, and computational analysis. Genomic privacy is primarily an informational harm, not a physical one.

2.2.2 Katz v. United States: The Reasonable Expectation of Privacy

In *Katz v. United States*, 389 U.S. 347 (1967), the Court held that the Fourth Amendment protects "people, not places," and that a search occurs when the government violates an individual's "reasonable expectation of privacy." Justice Harlan's concurrence articulated the two-pronged test: (1) the individual must have exhibited an actual (subjective) expectation of privacy, and (2) the expectation must be one that society is prepared to recognize as reasonable (objective).

Doctrinal significance for genomic privacy: The Katz framework is the doctrinal entry point for genomic privacy claims. An individual who provides a DNA sample to a healthcare provider, research institution, or consumer testing company retains a subjective expectation that the genomic information will be used only for the disclosed purpose. Whether society is prepared to recognize that expectation as reasonable depends, in part, on the five foundational characteristics: if the data is immutable, uniquely identifying, predictive, familial, and persistently sensitive, society has strong reasons to recognize the privacy expectation as reasonable even after limited disclosure.

2.2.3 The Third-Party Doctrine: Smith v. Maryland and United States v. Miller

The third-party doctrine holds that an individual retains no reasonable expectation of privacy in information voluntarily conveyed to third parties. *Smith v. Maryland*, 442 U.S. 735 (1979) (pen registers); *United States v. Miller*, 425 U.S. 435 (1976) (bank records). The rationale is that the individual "assumed the risk" that the third party would reveal the information to the government.

Doctrinal significance for genomic privacy: The third-party doctrine, if applied mechanically to genomic data, would extinguish constitutional protection for genomic information disclosed to healthcare providers, testing companies, or research institutions. This result is untenable for five reasons, each corresponding to a foundational characteristic:

1. **Immutability (CP-001):** Unlike financial records or telephone numbers, genomic information cannot be changed if the third-party custodian discloses it.
2. **Familial Impact (CP-004):** The third-party doctrine assumes that the individual who discloses information is the sole affected party. Genomic disclosure inherently affects biological relatives who made no disclosure.
3. **Persistent Sensitivity (CP-005):** The doctrine assumes that the scope of disclosed information is fixed at the moment of disclosure. Genomic information becomes more revealing over time.
4. **Predictive Capacity (CP-003):** The doctrine developed in contexts where disclosed information reflected past conduct or present status. Genomic information reveals probabilistic future states.
5. **Nature of Disclosure:** Disclosure to a third party for a limited purpose is not equivalent to disclosure to the government for unrestricted purposes.

2.3 Digital Privacy Doctrine: The Carpenter Revolution

2.3.1 Carpenter v. United States

In *Carpenter v. United States*, 585 U.S. __ (2018), the Court held that the government's acquisition of cell-site location information (CSLI) from wireless carriers constituted a search under the Fourth Amendment, notwithstanding the third-party doctrine. The Court identified several factors that rendered CSLI distinct from the records in *Smith* and *Miller*. Each factor maps precisely onto genomic data:

Carpenter Factor	Genomic Analog
Depth, breadth, comprehensive reach	Whole-genome sequence is the most comprehensive biological identifier; it reveals ancestry, health risks, traits, and familial relationships.
Automatic generation	Genomic sequence is biologically fixed, not chosen or generated by the individual.

Carpenter Factor	Genomic Analog
Indispensability	Participation in healthcare, research, or family medical history increasingly requires genomic disclosure.
Time-span	Genomic data covers the individual's entire lifetime and reveals information about past, present, and future biological states.

2.3.2 *Riley v. California*: Digital Containers and the "Privacies of Life"

In *Riley v. California*, 573 U.S. 373 (2014), the Court held that the search of a cell phone incident to arrest requires a warrant, emphasizing that digital devices contain "the privacies of life" in digital form. A genome, like a cell phone, contains "the privacies of life" in a form that is quantitatively vast and qualitatively distinct from physical items.

2.3.3 *United States v. Jones*: GPS Tracking and Prolonged Surveillance

In *United States v. Jones*, 565 U.S. 400 (2012), the Court recognized the "mosaic theory"—that aggregation of data points creates a privacy harm greater than the sum of its parts. A GPS device tracks location for a period of weeks; a genomic database tracks biological identity for a lifetime.

2.4 Government Collection of Genetic Information: *Maryland v. King*

In *Maryland v. King*, 569 U.S. 435 (2013), the Court upheld the collection of DNA from arrestees strictly because it was used "solely for identification" and relied only on non-coding markers. The five foundational characteristics challenge the *King* safe harbor when applied to whole genomes:

King Limitation	Genomic Data Challenge
Used solely for identification	Whole-genome data cannot be limited to identification; it inherently contains predictive and trait information.
Non-coding markers only	Whole-genome sequencing includes coding regions.
Minimal physical intrusion	The informational intrusion from genomic analysis far exceeds the physical intrusion of collection.
Arrestee context	Governmental genomic collection increasingly occurs in non-criminal contexts (research, healthcare, employment).

2.5 Informational Privacy: *Whalen v. Roe* and Its Progeny

Whalen v. Roe, 429 U.S. 589 (1977), recognized a constitutional "interest in avoiding disclosure of personal matters". Genomic information falls within the core of personal matters that require protection against unmonitored government storage or dissemination. While *NASA v. Nelson*, 562 U.S. 134 (2011), deferred to legislative frameworks for standard employment records, the profound qualitative features of genomic information necessitate an explicit judicial framework.

2.6 The Doctrinal Gap: Why Genomic Information Breaks Existing Categories

2.6.1 The Category Problem: Genomic data transcends all previous classifications by combining physical source, permanent biometric matching, and forward-looking probabilistic health forecasting into a single data type.

2.6.2 The Aggregation Problem: Computational integration of a genome with public registries creates a biological mosaic whose surveillance potential eclipses any single physical trace.

2.6.3 The Temporal Problem: Unlike standard records whose sensitivity degrades or remains fixed, a genome's latent information increases over time due to ongoing functional genomic discovery.

CHAPTER 3: Constitutional Analysis

3.1 Purpose

This chapter applies the proposed tiered scrutiny framework across concrete legal and administrative domains to establish its operational utility.

3.2 Criminal Investigations

3.2.1 Forensic Identification: Routine matching using limited, non-coding CODIS short tandem repeats (STRs) remains within Tier One (reasonableness review).

3.2.2 Phenotypic Inference and Genetic Genealogy: Investigating ancestry or using consumer databases (GEDmatch) falls under Tier Two (heightened scrutiny), requiring narrow tailoring, demonstrated exhaustion of regular leads, and explicit judicial warrant authorization.

3.2.3 Compelled Whole-Genome Disclosure: Mandating a suspect's complete sequence falls under Tier Three (strict scrutiny), sustainable only under supreme emergencies where identity is critical and no narrow method exists.

3.3 Administrative Benefits Programs

3.3.1 Social Security Disability Adjudication: Voluntary submission of genetic data to support a diagnosis maps to Tier One. However, if the state mandates genetic sequencing or predictive profiling as a condition of eligibility, Tier Three strict scrutiny is triggered.

3.3.2 Employment and Insurance Contexts: GINA bars health and workplace genetic discrimination but excludes life or long-term care insurance. When state agencies act as employers, Fourteenth Amendment due process demands heightened scrutiny for compelled disclosure.

3.4 Public Health Surveillance

Pathogen tracking that isolates viral signatures falls under Tier One. Long-term human data biobanks (e.g., All of Us Research Program) involve extensive record linkages, requiring Tier Two procedural oversight, strict purpose limitations, and cryptographic segregation.

3.6 Summary of Tiered Application

Context	Typical Tier	Key Trigger
Forensic identification (limited markers)	Tier One	Non-coding, non-predictive, sole-purpose identification.

Context	Typical Tier	Key Trigger
Phenotypic inference / genetic genealogy	Tier Two	Secondary analysis, familial implications, database aggregation.
Compelled whole-genome (criminal)	Tier Three	Comprehensive collection, predictive capacity, familial impact.
Disability adjudication (supplementary)	Tier One	Limited scope, consent-based, immediate diagnostic use.
Disability adjudication (mandatory profiling)	Tier Three	Compelled comprehensive disclosure, benefits conditioning.
Research (law enforcement access)	Tier Two	Aggregation, secondary use, absence of specific user consent.

CHAPTER 4: Question Presented

Whether whole-genome information constitutes a distinct category of personal information entitled to heightened constitutional protection under the Fourth and Fourteenth Amendments because genomic information is immutable, uniquely identifying, predictive of future biological conditions, inherently reveals information regarding biological relatives, and remains sensitive throughout an individual's lifetime.

Subsidiary Questions

1. Whether individuals retain a reasonable expectation of privacy in genomic information disclosed to third-party custodians.
2. Whether compelled disclosure, acquisition, retention, or analysis of genomic information implicates constitutional liberty or privacy interests distinct from those associated with ordinary medical records.
3. Whether existing constitutional doctrines adequately address the unique characteristics of genomic information.
4. Whether heightened procedural safeguards are warranted before governmental acquisition, access, transfer, retention, or secondary use of genomic information.

CHAPTER 5: Proposed Constitutional Framework

This framework does not invent a new right; it asserts that existing tests under *Katz*, *Carpenter*, and *Whalen* must formally incorporate the five physical realities of genomic data to remain doctrinally sound.

5.4 Tiered Scrutiny Model

Tier One: Limited Identification Uses

Reasonableness review applies if data is restricted to non-coding forensic markers used solely for matching, with zero trait extraction and immediate deletion parameters.

Tier Two: Heightened Scrutiny for Retention and Aggregation

Applies when data is aggregated or processed for secondary computational features (PRS, lineage matching). Requires an

articulated substantial state interest, strict narrow tailoring, and prior authorization by a neutral magistrate.

Tier Three: Strict Scrutiny for Compelled Whole-Genome Profiling

Triggered by state-compelled whole-genome capture or predictive biological staging for administrative inclusion/exclusion. Requires demonstration of a compelling governmental interest, narrow tailoring, and execution via the least restrictive means available.

5.5 Concrete Procedural Safeguards

5.5.1 Judicial Authorization: Warrants or necessity orders are structural prerequisites for Tier Two or Three access.

5.5.2 Purpose Limitations: Captured signatures cannot be dynamically mapped to alternate operational goals without a separate warrant loop.

5.5.3 Retention Limits: Persistent holding of human blueprints without discrete case closure parameters is presumptively unconstitutional.

5.5.6 Data Minimization: Capturing the whole sequence when a localized variant answers the state's inquiry fails tailoring analysis.

5.5.8 Secure Computation Environments: State-mandated analytics must be restricted to cryptographically isolated runtimes where raw sequence layers are hidden from human operators.

VOLUME II: GENOMIC SECURITY & GOVERNANCE

Executive Summary

Volume II translates the legal criteria of Volume I into precise infrastructure engineering requirements. It defines the technical standards for secure confidential computation, post-quantum cryptography, decentralized data tracking, and algorithmic auditing required to preserve biological privacy at scale.

PART A: SECURE ENCLAVE ARCHITECTURE

A.2 Threat Model

Table A.2.1: Threat Actors

Threat Actor	Capability	Target
External Attacker	Network intrusion, malware, credential theft.	Genomic database, variant files.
Insider (SysAdmin)	Privileged server memory and storage access.	Plaintext FASTA/VCF file layers, raw queries.
Cloud Provider	Physical access to host, hypervisor manipulation.	Active RAM, virtual disk images.

A.3.4 Cryptographic Remote Attestation Flow

The structural interaction sequence for enforcing judicial boundaries on secure runtime containers is detailed below:

- 1. Court Issues Authorization Token -> Cryptographically signed scope & hash constraints.
- 2. Enclave Boot & Measurement -> CPU computes SHA-256 code hash of loaded pipelines.
- 3. Challenge-Response Verification -> Independent verifier checks code signature + hardware RoT.

4. Plaintext Key Release -> Encrypted key released to isolated CPU cache only.
5. In-Enclave Decryption -> Data parsed within hardware-encrypted memory boundary.
6. Execution and Minimization -> Exits restricted strictly to aggregate scores; logs chained.

PART B: FEDERATED LEARNING SYSTEMS

Federated processing decentralizes training across distinct clinical biobanks. The master orchestrator handles model parameter optimization rounds but never gains visibility into the raw local sequence records, mitigating centralized breach exposure.

PART C: DIFFERENTIAL PRIVACY MODELS

To block background linkage and reconstruction attacks, all public data outputs must pass through mathematical noise introduction. For government research releases, strict composition accounting enforces cumulative privacy expenditure boundaries (Maximum $\epsilon \leq 0.5$ for active research sharing; $\epsilon \leq 1.0$ for macro public health indices).

PART D: CRYPTOGRAPHIC CONTROLS & KEY MANAGEMENT

Decryption master keys must reside within Hardware Security Modules (HSMs) managed under judicial escrow. Operational administrative entities cannot independently trigger decryption without generating an automated, unalterable ledger entry tied to a valid authorization token.

PART E: POST-QUANTUM SECURITY STANDARDS

Because human genomic identity remains persistent across generations, intercept-now-decrypt-later vectors threaten long-term security. All archival structures must migrate to quantum-resistant standards:

- **Key Encapsulation:** ML-KEM (FIPS 203) parameter set ML-KEM-768 for session establishing.
- **Digital Signatures:** ML-DSA (FIPS 204) parameter set ML-DSA-65 for authorization token verification.

VOLUME III: DISABILITY, ACCESSIBILITY & ADMINISTRATIVE SYSTEMS

Executive Summary

Volume III evaluates the evidentiary role of genetic profiling within administrative frameworks, primarily the Social Security Administration (SSA). It structures protections ensuring genomic insights remain strictly secondary and auxiliary, preventing structural exclusion or automated diagnostic profiling of vulnerable individuals.

PART A: DISABILITY ADJUDICATION OVERVIEW

Under the five-step sequential evaluation path (20 C.F.R. § 404.1520), genomic profiles may only serve as supplementary medical evidence at Step 2 (Severity evaluation) or Step 3 (Listings matching) to confirm a diagnosis. A genetic score or variant classification, on its own, is fundamentally incapable of documenting functional impairment or the inability to execute substantial gainful activity.

PART B: ACCESSIBILITY & NOTICE STANDARDS

In alignment with Section 508 of the Rehabilitation Act and WCAG 2.1 Level AA, all genomic disclosure frameworks and privacy notices must be deployed in accessible, screen-reader compatible layouts, clear plain language (6th-grade maximum baseline), or alternate tactile formats upon claimant request.

PART G: GOV-LLM ACCESSIBILITY SAFEGUARDS

Large Language Models deployed for citizen assistance must never execute training rounds or fine-tuning weights using identifiable genomic inputs. Public-facing analytical queries mapping hereditary terminology must be executed inside secure hardware enclaves using strict privacy-preserving zero-retention parameters.

VOLUME IV: SCIENTIFIC RECORD & VALIDATION

Executive Summary

Volume IV anchors the portfolio's legal and security architectures within established bioinformatics standards. It mandates reproducibility pipelines, explicit pipeline quality control thresholds, and objective interpretation rubrics, framing genomic metrics strictly as scientific inputs rather than final legal conclusions.

PART A: BIOINFORMATICS METHODOLOGIES

All public or administrative variant pipeline workflows must rely on containerized execution platforms (Docker/Singularity) pinned to explicit build iterations. The Human Build 38 (GRCh38) configuration is the mandatory baseline reference standard for all constitutional or forensic processing.

Pipeline Stage	Standard Tooling	Quality Control Metric
Read Alignment	BWA-MEM, Minimap2	Mapping quality scores; uniform target coverage depth.
Variant Calling	GATK HaplotypeCaller, DeepVariant	Variant Quality Score Recalibration (VQSR) filtering.

PART C: VARIANT INTERPRETATION FRAMEWORKS

Interpretation models must rigidly execute the ACMG/AMP consensus standards. Variants of Uncertain Significance (VUS - Class 3) are strictly inadmissible as sole objective proof of an administrative or physiological impairment claim.

PART D: POLYGENIC RISK SCORE LIMITATIONS

Polygenic Risk Scores represent mathematical probabilities, not discrete diagnoses. Because current training datasets suffer from extreme population stratification (skewed heavily toward European ancestry cohorts), using standalone PRS indexes inside public allocation programs or adjudication reviews introduces systemic inequities and is strictly prohibited.

VOLUME V: LEGISLATIVE & REGULATORY FRAMEWORKS

Executive Summary

Volume V codifies the constitutional principles of Volume I and the software targets of Volume II into statutory text, drafting a comprehensive model statute to govern biological data privacy.

PART A: FEDERAL GENOMIC PRIVACY ACT PROPOSAL (FGPA)

Be it enacted by the Senate and House of Representatives of the United States of America in Congress assembled,

SECTION 1. SHORT TITLE.

This Act may be cited as the "Federal Genomic Privacy Act of 2026."

SECTION 4. TIERED PROTECTION STANDARDS.

- (a) *Tier One (Forensic Linkage)*: State actors may capture localized non-coding identification markers without a warrant, provided no secondary phenotypic forecasting occurs.
- (b) *Tier Two (Retained/Aggregated Structures)*: Retention or secondary processing demands prior judicial authorization based on an explicitly proven substantial state interest and narrow tailoring.
- (c) *Tier Three (Compelled Whole-Genome Capture)*: Compelling complete germline disclosure or profiling for administrative eligibility is unlawful unless justified by a compelling state interest and executed via the least restrictive means.

SECTION 6. STATUTORY LIMITS ON THE THIRD-PARTY DOCTRINE.

Citizens retain a clear, reasonable expectation of privacy in genomic data held by commercial, research, or clinical custodians. The third-party exceptions created under *Smith* and *Miller* do not apply to biological sequence records.

SECTION 7. REMEDIES & THE EXCLUSIONARY RULE.

Provides an explicit private right of action for civil damages. Any genomic profile captured or analyzed in violation of this statute is strictly inadmissible across all federal criminal, civil, or administrative venues.

MASTER AUTHORITY MATRIX

The complete cross-reference directory mapping legal claims to technical design targets is tabulated below:

Authority ID	Core Proposition	Structural Source	Exhibit
CP-001	Genome is immutable; compromise is completely irreversible.	Vol. I, § 1.3.1; § 5.2.1	EX-0001
CP-002	Genome is uniquely identifying; anonymization is mathematically fragile.	Vol. I, § 1.3.2; § 5.2.2	EX-0002
CP-004	Genomic disclosure directly compromises non-consenting biological kin.	Vol. I, § 1.3.4; § 5.2.4	EX-0004
CP-007	<i>Carpenter</i> surveillance factors map directly onto deep genomic structures.	Vol. I, § 2.3.1	EX-0006
CP-010	<i>King</i> limits protection strictly to forensic matching; bars broader data use.	Vol. I, § 2.4.1; § 3.2.1	EX-0007
SG-001	Secure enclaves enforce isolated pipeline processing and protect active RAM.	Vol. II, Part A, § A.1	EX-0010
SG-003	Remote attestation checks runtime configurations before releasing data keys.	Vol. II, Part A, § A.3.4	EX-0011

Authority ID	Core Proposition	Structural Source	Exhibit
SG-010	Post-quantum primitives protect archival stores against future cryptanalysis.	Vol. II, Part E	EX-0016
DA-001	Genomic variants are auxiliary inputs; they cannot independently verify disability.	Vol. III, Part A, § A.3	EX-0017
SR-003	ACMG/AMP sequence rules represent the mandatory evaluation framework.	Vol. IV, Part C, § C.2	EX-0019
LF-001	Model legislation establishes durable statutory protections across federal networks.	Vol. V, Part A, § A.2	EX-0022

VERSION CONTROL

Version	Date	Author	Changes
1.0	2026-06-24	Portfolio Architect	Complete five-volume production compilation; populated authority mapping metrics.

* END OF PORTFOLIO DOCUMENT *